

In-water clean and biofouling inspection of the *MV Example* to demonstrate compliance with New Zealand and Australian biofouling regulations

Initial Inspection: 19/09/18: Singapore Cruise Centre, Singapore
In-water clean: 17/11/18: Laem Chabang, Thailand.
Final inspection: 23/11/18: Singapore Cruise Centre, Singapore

Prepared for **Company X**



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EXECUTIVE SUMMARY

Company X is managing the 221 m cruise ship *MV Example* on behalf of X Cruises. In the 2018-19 cruise season the vessel will operate in the waters of Australia, Southeast Asia, Asia and New Zealand. Given the *MV Example* will visit several jurisdictions in Australia and New Zealand where management of biofouling or Invasive Marine Species (IMS) is a regulatory requirement, Company X has engaged the services of X to conduct an in-water biofouling inspection and oversee cleaning operations to assist in managing any observed biofouling to reduce the risk of transferring IMS of concern to destination ports. The biofouling inspection and cleaning operations were conducted while the *MV Example* was alongside at the Singapore Cruise Centre, Singapore on 19 September, 2018; while the vessel was at anchor in Laem Chabang, Thailand on 17 November, 2018; and while alongside at the Singapore Cruise Centre, Singapore on 23 November, 2018

An initial in-water biofouling inspection of the *MV Example* conducted while alongside at Singapore Cruise Centre, Singapore on 19 September, 2018, determined that the hull was in good condition with biofouling on the general hull typically consisting of diatomaceous slime and a waterline band of green algae. Secondary biofouling on the hull plating was largely restricted to the forward sections of the waterline, the aft sections of waterline and the starboard quarter of the transom. This secondary biofouling was associated with a band of green algae extending to approximately 2.5 m depth and consisted of scattered barnacles (*Amphibalanus reticulatus* <5%) and tubeworms (*Hydroides elegans* <1%), with biofouling concentrated on weld seams and the waterline. Some scattered secondary biofouling consisting of individual acorn barnacles and tubeworms (<1%) was detected over other areas of the hull plating. A thorough inspection of niche areas determined that biofouling in niche areas was typically restricted to secondary biofouling composed of acorn barnacles (c.f. *A. reticulatus*), tubeworms (*H. elegans*), hydroids and bryozoans. In most cases biofouling in niche areas represented less than 5% coverage. However, biofouling associated with overboard pipes, draft markers (forward and aft), tunnel thruster propeller blades, anodes associated with the port stabiliser fin, rope guard voids and rudder stock areas exceeded this level. Localised secondary biofouling in some restricted niche areas represented up to 90% coverage. Tertiary biofouling composed of scattered individual mussels (*Mytilus galloprovincialis*) was restricted to the rudder stock areas and a single individual associated with a port rope cutter blade.

Based on the biofouling observed in the initial inspection, X designed a scope of work to remove all viable biofouling in excess of the New Zealand Craft Risk Management Standard (NZ CRMS) short-stay threshold. This scope of work was completed through two dive operations overseen by a X inspector with the first occurring in Laem Chabang, Thailand on 17 November, 2018, and the second occurring while alongside at the Singapore Cruise Centre on 23 November, 2018. Divers used a range of techniques to remove

biofouling including a diver operated brush kart targeting waterline biofouling, high pressure water jets operated at 3000 psi to remove biofouling from complex or curved niche areas such as rope guard voids and overboard pipes, and hand scrapers to remove biofouling from accessible niche areas. Tunnel thrusters were accessed via SCUBA divers through the fixed grating and all surfaces were hand cleaned to remove biofouling.

The final dive operation in Singapore on 23 November, 2018 included a full biofouling inspection to ensure that all cleaning was conducted to a high standard, and that no biofouling in excess of the NZ CRMS short-stay threshold persisted on the submerged surfaces and immersible equipment of the *MV Example*. The X inspector also confirmed that no macroscopic IMS of concern listed in Australian state regulation had recruited to the hull since the initial inspection was conducted on 19 September, 2018. As such, at the conclusion of the final IMS inspection the *MV Example* was found to be in compliance with all regulatory obligations associated with biofouling during the documented 2018/2019 cruise season.

X acknowledges that microscopic life-history stages may have been present in association with the *MV Example* and not detected, and that individuals of IMS of concern or other biofouling species may settle on the vessel's submerged hull subsequent to the inspection. To minimise the chance of infection with IMS of concern, the *MV Example* should limit exposure to IMS propagules by minimizing time spent in coastal waters prior to entering Australian waters. Considering that the *MV Example* is not scheduled to enter New Zealand waters until 21 January, 2019 (~58 days from the final inspection) there is a considerable potential for additional biofouling to accumulate on the submerged surfaces prior to entry. Though a range of factors may contribute to the rate of biofouling accumulation, there is a real possibility that by January 2019, biofouling in some areas of the *MV Example* may exceed the NZ CRMS short-stay threshold. However, due to regulations on in-water cleaning in Australia no further opportunities to remove biofouling prior to entry to New Zealand are available. X considers that the measures described here represent best practice in reducing biosecurity risks to New Zealand associated with biofouling and the measures adopted by the *MV Example* are consistent with the principal of reducing biosecurity risks to as low as reasonably practicable (ALARP).

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1. INTRODUCTION

1.1 Invasive Marine Species Background

Invasive Marine Species (IMS) are widely recognised as one of the most significant threats to marine ecosystems worldwide, along with overfishing and human-driven atmospheric and oceanic alterations. Shallow coastal marine environments in particular, are thought to be amongst the most heavily invaded ecosystems on earth, which largely reflects the accidental transport of IMS by international shipping. A number of human-mediated vectors can transport IMS outside their native ranges. For instance ocean-going vessels can transport marine species in ballast water, as biofouling attached to submerged immersible equipment, within internal seawater systems and/or on the exterior of the hull. For the past three decades, ballast water was considered the primary vector responsible for the dispersal of IMS around the world (Carlton 1985; Thresher et al. 1999; Eldredge and Carlton 2002). However, recent research suggests the role of ballast water was probably overstated, and up to 69% of these introductions may have occurred via biofouling (Hewitt et al. 1999, 2004; Hewitt and Campbell 2010). To prevent further introductions, and limit the spread of IMS, shipping requires management for both ballast water and biofouling.

Around the globe there are approximately 2,000 non-indigenous species that are known to have been introduced to marine or estuarine systems. Some of which have had catastrophic ecological, economic, human health and social/cultural consequences (Carlton 1996, 2001; Pimental et al. 2000; Hewitt et al. 2011). In Australian marine waters there are approximately 450 established marine species that are either non-indigenous, or whose origins are inconclusive (Hewitt et al. 2011). IMS that survive and establish reproductive populations in regions where they did not previously occur, often have no known impacts, although some species can cause major damage and are referred to as 'marine pests' or 'IMS of concern'. Some IMS can facilitate the introduction of additional species and their cumulative impacts can significantly undermine the structure, function and persistence of native ecosystems. It is predicted that each year approximately three to four additional IMS will continue to establish in Australian waters unless marine biosecurity management improves.

Once established, IMS are often difficult or impossible to eradicate. Most attempts to eradicate IMS are logistically challenging, complex, costly and unsuccessful. The pragmatic approach to protecting marine environments from the impacts of IMS, is to prevent their introduction and spread by human-mediated transport. This approach to marine biosecurity provides the fundamental rationale for most global and regional management initiatives.

2. BIOFOULING AND IMS MANAGEMENT REGULATIONS RELEVANT TO THE *MV EXAMPLE*

2.1 The *MV Example*

Company X is managing the 221 m cruise ship *MV Example* on behalf of X Cruises. In the 2018-19 cruise season the vessel will operate in the waters of Australia, Southeast Asia, Asia and New Zealand. Given the *MV Example* will visit several jurisdictions where management of biofouling or Invasive Marine Species (IMS) is a regulatory requirement, Company X has engaged the services of X to conduct an in-water biofouling inspection of the external hull, and to assist in managing any observed biofouling to reduce the risk of transferring IMS of concern to destination ports. Furthermore, Company X requested that X offer advice on what diver-cleaning measures can be adopted during subsequent port stays to remove any excessive biofouling.

An initial biofouling inspection was conducted while the *MV Example* was alongside at the Singapore Cruise Centre, Singapore on 19 September, 2018. Following the identification of biofouling in excess of regulatory requirements in New Zealand, X then developed a scope of work for remedial cleaning which was subsequently conducted in Laem Chabang, Thailand on 17 November, 2018, and while alongside at the Singapore Cruise Centre, Singapore on 23 November, 2018. This report documents the outcome of the inspection and cleaning operations and provides advice to Company X on the best approach to manage the vessel to demonstrate compliance with regulatory obligations.

2.2 Relevant Biofouling Regulations

Subsequent to the IMS inspection the *MV Example* is scheduled to visit a number of destination ports in Asia, Australia and New Zealand. A summary of destination ports between September 2018 and March 2019 is provided in **Figure 1** with ports marked by a star indicating locations with mandatory regulations in place providing authorities with powers in relation to either the presence of specific IMS (i.e. Australia) or biofouling beyond a specific threshold (i.e. New Zealand). Given the intended itinerary of the *MV Example*, three key biofouling management obligations need to be considered:

- 1) Mandatory compliance with state legislation in Australia.
- 2) Mandatory compliance with the New Zealand *Biosecurity Act 1993* and the Craft Risk Management Standard (CRMS) for biofouling.



Figure 1 A summary of key destination ports for the *MV Example* in the 2018-2019 cruise seasons. Destinations marked with a ★ indicate ports subject to Australian state regulatory obligations governing the import of listed “noxious” aquatic species (see Section 2.3). Destinations marked with a ★ indicate ports subject to the Craft Risk Management Standard governing acceptable thresholds for biofouling (see Section 2.4).

2.3 Australian Marine Biosecurity Management

A key role of the Commonwealth's Department of Agriculture and Water Resources (DAWR) is to safeguard Australia's favourable human, animal and plant health status. Previously, this was regulated under the *Quarantine Act 1908* and its subordinate legislation (i.e. *Quarantine Regulations 2000* and the *Quarantine Proclamation 1998*). However, these have now been superseded by the *Biosecurity Act 2015* which was introduced on 16 June, 2016. The key legislative change between the *Quarantine Act 1908* and the *Biosecurity Act 2015* is the jurisdictional shift of the DAWR from 200 nm to 12 nm (e.g. Australian territory).

At this stage there are no intentions within the *Biosecurity Act 2015* to manage biofouling on facilities or vessels entering Australian territory. However, despite there being no mandatory Commonwealth biofouling regulations in place, the Commonwealth previously released guidance documents for national biofouling management as part of the 'national system for the prevention and management of marine pest incursions' covering various vessel classes and industry sectors. These notably include guidelines for the petroleum production and exploration industry. For further details see (<http://marinepests.gov.au/petroleum/managing-biofouling>).

Considering the lack of specific biofouling management regulations at the Australian Commonwealth level, states and Territories within Australia have developed a range of regulatory tools to protect their state waters from the threats associated with IMS. While all States and Territories of Australia have some powers to act against the introduction of IMS of concern, some States/Territories are more active than others, and in some instances legislation that implies specific powers to state authorities are not actively enforced. A review of biofouling species specifically prohibited in state water that will be visited by the *MV Example* are summarised in **Table 1** along with a reference to the relevant regulatory mechanisms. In all cases these powers are confined to state waters (i.e. within 3 nm seaward of the Territorial Sea baseline).

It should be noted that no specific regulations regarding vessel inspections are in place in any of the states to be visited by the *MV Example*. Thus while state authorities have the power to act in relation to a verified biosecurity risk, there is presently no formalised reporting process or compliance checks implemented.

Table 1 “Noxious” or “Prohibited” biofouling species listed under relevant Australian state legislation. NT Species: Noxious fish refer to those listed in Schedule 1 of the *Fisheries Regulations 2012*; QLD Species: Prohibited matter species are listed in Schedule 1 of the *Biosecurity Act 2014*; NSW Species: Prohibited matter listed in Schedule 2 of the *Biosecurity Act 2015*. VIC Species: Noxious Aquatic Species declared under Section 75 of the *Fisheries Act 1995* – updated list provided online: <https://vfa.vic.gov.au/operational-policy/pests-and-diseases/noxious-aquatic-species-in-victoria> (last updated 2010); TAS species: Noxious Fish listed in Part 4, Section 20 of *Fisheries (General and Fees) Regulations 2006*.

Phylum	Scientific Name	Common Name	NT	QLD	NSW	VIC	TAS
Algae	<i>Caulerpa taxifolia</i> (exotic strains only)	Green macroalga	x	x		x	x
Algae	<i>Codium fragile fragile</i>	Green macroalga	x	x			
Algae	<i>Grateloupia turuturu</i>	Red macroalga	x	x	x		
Algae	<i>Sargassum muticum</i>	Asian seaweed	x	x	x		
Algae	<i>Undaria pinnatifida</i>	Japanese seaweed	x	x		x	x
Annelida	<i>Alitta succinea</i>	Pile worm	x				
Annelida	<i>Boccardia proboscidea</i>	Spionid polychaete	x				
Annelida	<i>Euchone limnicola</i>	Sabellid polychaete worm	x				
Annelida	<i>Marenzelleria</i> spp.	Red gilled mudworm	x	x			
Annelida	<i>Polydora websteri</i>	Mudworm	x				
Annelida	<i>Polydora cornuta</i>	Spionid polychaete	x				
Annelida	<i>Sabella spallanzanii</i>	European fan worm	x	x		x	x
Ascideacea	<i>Ciona intestinalis</i>	Sea vase	x				
Ascideacea	<i>Didemnum</i> spp. (exotic invasive strains only)	Colonial sea squirt	x	x			
Ascideacea	<i>Didemnum vexillum</i>	Colonial sea squirt	x	x	x		
Ascideacea	<i>Styela clava</i>	Clubbed tunicate	x				
Cnidaria	<i>Mnemiopsis leidyi</i>	Comb jelly	x	x	x		

Phylum	Scientific Name	Common Name	NT	QLD	NSW	VIC	TAS
Crustacea	<i>Balanus improvisus</i>	Barnacle	x	x	x		
Crustacea	<i>Eriocheir</i> spp.	Chinese mitten crab	x	x	x		
Crustacea	<i>Hemigrapsus sanguineus</i>	Japanese/Asian shore crab	x	x	x		
Crustacea	<i>Petrolisthes elongatus</i>	New Zealand half shell crab	x				
Crustacea	<i>Hemigrapsus takanoi/penicillatus</i>	Pacific crab	x	x	x		
Crustacea	<i>Charybdis japonica</i>	Lady crab	x	x	x		
Crustacea	<i>Carcinus maenas</i>	European green crab	x	x			x
Crustacea	<i>Procambarus clarkii</i>	Red swamp crayfish			x	x	
Echinoderm	<i>Asterias amurensis</i>	Northern Pacific seastar	x	x	x	x	x
Teleost	<i>Neogobius melanostomus</i>	Round goby	x	x	x		
Teleost	<i>Siganus rivulatus</i>	Marbled spinefoot, rabbit fish	x	x	x		
Holoplankton	<i>Alexandrium monilatum</i>	Toxic dinoflagellate	x	x			
Holoplankton	<i>Dinophysis norvegica</i>	Toxic dinoflagellate	x	x			
Holoplankton	<i>Pfiesteria piscicida</i>	Toxic dinoflagellate	x	x			
Holoplankton	<i>Chaetoceros concavicornis</i>	Centric diatom	x	x			
Holoplankton	<i>Chaetoceros convolutus</i>	Centric diatom	x	x			
Holoplankton	<i>Pseudo-nitzschia seriata</i>	Pennate diatom	x	x			
Mollusca	<i>Musculista senhousia</i>	Asian bag mussel	x	x	x		
Mollusca	<i>Mytilopsis sallei</i>	Black striped mussel	x	x	x	x	x
Mollusca	<i>Perna perna</i>	Brown mussel	x	x	x		
Mollusca	<i>Perna viridis</i>	Asian green mussel	x	x	x		
Mollusca	<i>Perna canaliculus</i>	New Zealand green mussel			x		

Phylum	Scientific Name	Common Name	NT	QLD	NSW	VIC	TAS
Mollusca	<i>Corbula (Potamocorbula) amurensis</i>	Asian clam, brackish-water corbula	x	x	x		
Mollusca	<i>Ensis directus</i>	Jack-knife clam	x	x	x		
Mollusca	<i>Mya arenaria</i>	Soft shell clam	x	x	x		
Mollusca	<i>Theora lubrica</i>	Asian semelid bivalve	x				
Mollusca	<i>Varicorbula gibba</i>	European clam	x	x	x		
Mollusca	<i>Crepidula fornicata</i>	American slipper limpet	x	x	x		
Mollusca	<i>Rapana venosa (syn Rapana thomasi)</i>	Rapa whelk	x	x	x		
Mollusca	<i>Magalana gigas</i>	Pacific Oyster		x			
Mollusca	<i>Maoricolpus roseus</i>	New Zealand screw shell	x	x			

2.4 New Zealand Biofouling Management and the Craft Risk Management Standard

The biosecurity threat to New Zealand coastal waters posed by IMS entering territorial waters in association with biofouling is presently regulated via *Biosecurity Act 1993*. Specifically, Section 33(1)(b) of the Act states that:

- (1) *Where there are any risk goods on board or attached to the outside of a craft that has entered New Zealand territory from outside New Zealand territory, an inspector may direct the master or other person in charge of the craft to ...*
- (b) *move the craft outside New Zealand territory (immediately, or within a period specified by the inspector);*

As such, the identification of a vessel with biofouling containing any non-indigenous marine species considered as posing a risk to the New Zealand's marine resources, economy, environment, and/or people's health the New Zealand Ministry for Primary Industries (NZ MPI) has the power to deny the vessel entry into New Zealand's territorial waters or turn the vessel around and order its immediate departure.

Though the *Biosecurity Act 1993* provides broad powers for New Zealand to act upon identified biofouling related threats, the NZ MPI developed their Craft Risk Management Standard (CRMS) to better protect the waters of New Zealand from the threat posed by non-indigenous species arriving via biofouling. The CRMS sets clear standards for the management of biofouling risk associated with all vessels (sea craft) that are entering New Zealand territorial waters (within 12 nm from the coastal baseline). These standards represent the most comprehensive management regime for biofouling related introductions applied to any state in the world¹. The CRMS was designed with the intention to compliment the 2011 IMO Biofouling Guidelines. As such, the CRMS adopts an approach of managing the general level of biofouling as opposed to a species-specific approach implemented by other jurisdictions such as those in Western Australia and other regions.

In essence, the CRMS will require that all vessels visiting New Zealand territorial waters have a "clean hull", with the acceptable thresholds for this status being defined according to the intended length of stay in New Zealand territorial waters. Two categories of vessels are distinguished in the CRMS:

- 1) **'Long-stay vessels'** means those vessels intending to remain in New Zealand for 21 days or longer, or those vessels intending to visit areas other than those designated under Section 37 of the *Biosecurity Act 1993* as 'Places of First Arrival'.
- 2) **'Short-stay vessels'** means those vessels intending to remain in New Zealand for 20 days or less and to only visit places designated under section 37 of the Act as 'Places of First Arrival'. These

¹ Note that some localised high value areas such as Barrow Island in Western Australia, the Hawaiian National Monument and the New Zealand Subantarctic Islands are managed to similar or more stringent standards.

vessels generally remain under ‘biosecurity surveillance’ while in New Zealand territory rather than becoming fully cleared of risk goods.

Thresholds for the definition of a “clean hull” for each of these vessel categories are provided in **Table 2**.

Table 2. Allowable biofouling threshold relative to intended length of stay in New Zealand waters.

Length of stay	Hull Area	Allowable Biofouling
0 - 20 days (Short -stay)	All hull surfaces	Slime layer; Gooseneck barnacles.
	Wind and water line	Algae growth of unrestricted cover and no more than 50 mm in frond, filament or beard length; Brown and red algal growth of no more than 4 mm in length; Incidental (maximum of 1%) coverage of one organism type of either tubeworms, bryozoans or barnacles, occurring as: - Isolated individuals or small clusters; and - A single species, or what appears to be the same species.
	Hull area	Algal growth occurring as: - No more than 4 mm in length; and - Continuous strips and/or patches of no more than 50 mm in width. Incidental (maximum of 1%) coverage of one organism type of either tubeworms, bryozoans or barnacles, occurring as: - Isolated individuals or small clusters that have no algal overgrowth; and - A single species, or what appears to be the same species.
	Niche areas	Algal growth occurring as: - No more than 4 mm in length; and - Continuous strips and/or patches of no more than 50 mm in width. Scattered (maximum of 5%) coverage of one organism type of either tubeworms, bryozoans or barnacles, occurring as: - Widely spaced individuals and/or infrequent, patchy clusters that have no algal overgrowth; and - A single species, or what appears to be the same species; and Incidental (maximum of 1%) coverage of a second organism type of either tubeworms, bryozoans or barnacles, occurring as: - Isolated individuals or small clusters that have no algal overgrowth; and - A single species, or what appears to be the same species.
>20 days (long-stay)	All hull surfaces	Slime layer; Gooseneck barnacles.

The CRMS sets out three Acceptable Measures that are considered as sufficient for meeting the “clean hull” requirement:

- 1) Cleaning before visit to New Zealand**, (or immediately on arrival in a facility or by a system, approved by NZ MPI). All biofouling must be removed from all parts of the hull and this must be carried out less than 30 days before arrival to New Zealand or within 24 hours after time of arrival.
- 2) Continual Maintenance using best practice** including: application of appropriate antifoul coatings; operation of Marine Growth Prevention Systems (MGPS) within sea-chests; and in-water inspections with biofouling removal as required. Following the 2011 IMO Biofouling Guidelines is recognised as an example of best practice.
- 3) Application of Approved Treatments.** Treatments are approved and listed under the Approved Biosecurity Treatments MPI-STD- ABTRT.

Additional information regarding the three acceptable measures is provided in the draft Guidance Document for the CRMS - Biofouling on Vessels Arriving to New Zealand. Furthermore, as an alternative to the Acceptable Measures above, a vessel operator may submit, for NZ MPI approval, a Craft Risk Management Plan (CRMP) under Section 24K of the *Biosecurity Act 1993* (which includes steps that will be taken to reduce risk to the equivalent degree as meeting the requirements of this standard).

To demonstrate that an operator has effectively implemented one of the Acceptable Measures to meet the “clean hull” requirement, operators will be required to submit to NZ MPI the following information:

- 1) Prior to arriving in New Zealand all operators of vessels must provide to MPI the following information (via the Advanced Notice of Arrival and associated documents).*
 - a. Intended length of stay within New Zealand territory and intentions in respect of places to be visited;*
 - b. Whether the vessel has spent any extended periods mainly stationary in a single location;*
 - c. If the vessel is coming in to undergo biofouling cleaning on arrival, any formal arrangement for cleaning or treatment and whether they will be undertaken immediately upon arrival (within 24 hours);*
 - d. What measures have been or will be used to meet the requirements of the standard; or*
 - e. Whether the operator has chosen to operate an NZ MPI approved Craft Risk Management Plan (CRMP) as an alternative to meeting the requirements of the standard.*
- 2) The following information (if relevant) must be held on the vessel and provided to MPI in an appropriate form if requested. (This is in addition to information to be provided prior to arrival).*
 - a. Information on the antifouling regime and any marine growth prevention systems used;*

- b. Whether applying the IMO Biofouling Guidelines, including employing a biofouling management plan showing the hull maintenance and inspection regime and records of biofouling management kept, preferably consistent with the template in the IMO guidelines;*
- c. If applicable to the vessel, its latest International Anti-fouling System Certificate or International Anti-fouling System Declaration;*
- d. Date and reporting from the latest hull biofouling inspection.*

2.5 Craft Risk Management Plan

Cruise vessels are typically considered as “long stay” vessels under the CRMS as their itineraries include areas other than those designated under Section 37 of the Biosecurity Act 1993 as ‘Places of First Arrival’. While cruise vessels could ostensibly claim compliance with the CRMS through evidence of “continual maintenance using best practice” (i.e. Acceptable Measure 2), in practice such an approach is unlikely to maintain biofouling within the “long-stay” threshold. Furthermore, any compliance activities undertaken by NZ MPI that detect unacceptable biofouling as defined in the CRMS would likely result in significant regulatory penalties if this approach was adopted. MPI has formally advised the cruise industry that:

“After May 2018, a vessel which does not meet the “clean hull” threshold for its length of stay and itinerary will face action to manage the associated biosecurity risk. For cruise vessels this may include restricted itinerary or revoked approval to visit certain ports (i.e. Milford Sound, or other locations that aren’t approved as ports). Any expenses associated with compulsory cleaning or disruptions to a vessel’s schedule must be met by the vessel owner or operator”.

In the CRMS Recommendations for Cruise Vessels, the NZ MPI acknowledges that for cruise vessels in particular, it may not be feasible to inspect and/or clean the hull 30 days prior to every arrival in New Zealand (i.e. Acceptable Measure 1). In lieu of meeting the CRMS ‘long-stay’ threshold this way, NZ MPI encourages cruise vessel operators to develop a CRMP that outlines alternative ways that the vessel operator will manage biofouling. CRMPs can be approved on a vessel-by-vessel basis, or at the organisation or industry level. The CRMP would need to be compliant with Section 24K of the Biosecurity Act 1993 and receive the approval from the Director-General who may also indicate the period for which a CRMP is considered valid. Throughout the period that the plan has the approval of the Director-General, a person who complies with the plan for an activity to which the plan applies does not have to comply also with the applicable Craft Risk Management Standard for the activity.

The NZ MPI Guidance Document for the Development of a CRMP for vessels (Guidance Document) provides specific advice for the cruise industry regarding how to draft and apply for a CRMP. In particular, the document recognises that many cruise lines already use best practice when maintaining their vessels’ hulls. As such, NZ MPI advises that a CRMP should not entail much more than what most well-maintained

cruise vessels do already. For example, “a vessel may propose a plan that entails maintaining their hull through best practice, plus some additional maintenance before beginning the cruise season in New Zealand”. Furthermore, for cruise vessels entering under a “long-stay” requirement, MPI suggest a pragmatic reduction of cleaning obligations to cleaning any part of the hull fouled in excess of the “short-stay” threshold. CRMPs are designed to be flexible, and can include any combination of activities, as long as these are sufficient to manage biofouling to NZ MPI’s standards.

MPI have indicated to X on 14 September, 2018 that there have been some recent legal decisions within the Department that have raised the bar for documenting our approvals and as a result MPI has been going through a review of all their processes. For Cruise Vessels this means all risk goods associated with cruise vessels need to be formally managed under a Craft Risk Management Plan (CRMP). While the focus of MPI has largely been on biofouling they also require assurance for topside (terrestrial) biosecurity risks. However topside compliance should be easy to meet as the conditions are essentially no different to what all cruise lines are already doing, so no changes to current process or any additional documentation will be required. The standard processes and verification from MPI Border Clearance Services will occur but no extra documentation for topsides are required. For biofouling management, cruise vessels will need to supply a biofouling management plan within their CRMP application, and then comply with the measures in the CRMP by supplying a record book and inspection report prior to arrival.

3. VESSEL PARTICULARS AND BIOFOULING MANAGEMENT FOR THE *MV Example*

3.1 Vessel Details

OWNER:	Company X
CONTACT PERSON:	X
VESSEL/STRUCTURE'S NAME:	<i>MV Example</i>
FLAG:	X
PORT OF REGISTRY:	X
VESSEL/STRUCTURE TYPE:	Cruise Ship
IMO NUMBER:	X
CALL SIGN:	X
MARITIME MOBILE SERVICE IDENTITY (MMSI):	X
DISPLACEMENT (TONNES):	47,861
LENGTH (M):	221
BREADTH (M):	34
MAXIMUM DRAFT (M):	6.7
GENERAL ARRANGEMENT:	Provided in Appendix 1.

3.2 Antifouling coating information

DATE OF APPLICATION:	25 September, 2017
TIME ELAPSED SINCE ² :	420 Days (~14 months)
COATING TYPE:	Jotun SeaQuantum Ultra S
LIFE EXPECTANCY:	60 months
LOCATION OF APPLICATION:	All submerged surfaces
Antifouling Coating certification is included in Appendix 2.	

² "Time elapsed since" is the period of time between the previous dry-dock/antifoul or inspection, and the current final inspection.

4. BIOFOULING CLEANING AND INSPECTION: METHODS AND RESULTS

4.1 Initial External Hull Inspection

X conducted an initial biofouling inspection operation while the *MV Example* was alongside at the Singapore Cruise Centre, Singapore on 19 September, 2018. Following the detection of biofouling in excess of the NZ CRMS “short-stay” threshold, the findings of this inspection were used to define a scope of work for subsequent cleaning operations. A summary of inspection methodology and outcomes for the initial inspection is provided below:

START DATE AND TIME:	11:50 hours, 19 September, 2018	
FINISH DATE AND TIME:	17:00 hours, 19 September, 2018	
DIVE COMPANY:	X	
DIVE VESSEL:	X	
DIVE SUPERVISOR:	X	
DIVER:	X (Dive 1), X (Dive 2)	
DIVING AND INSPECTION EQUIPMENT USED:	☒ Surface supply (SSBA) ☒ Digital Camera	☒ Communications and CCTV ☒ Scraper and catch-bag
ALL HULL LOCATIONS/NICHE AREAS INSPECTED:	☐ Yes	☒ No
HULL LOCATIONS/NICHE AREAS NOT INSPECTED:	Sea chests and tunnel thruster inspected through gratings only. Only a selection of overboard discharge pipes inspected. Stabiliser fins inspected in the retracted position.	
WATER CLARITY DURING INSPECTION:	2 m	
TIDAL CURRENTS DURING INSPECTION:	<1 knots	
INSPECTION RECORDED ON DVD:	☒ Yes	☐ No
Comments: See X (2018a)³ for detailed results. The initial inspection determined that the hull of the <i>MV Example</i> was generally clean with biofouling limited to acorn barnacles (<i>Amphibalanus reticulatus</i>) and serpulid tubeworms (<i>Hydroides elegans</i>). Isolated mussels (<i>Mytilus galloprovincialis</i>) were also detected in association with the rudder and rope guards of the main propeller. No species detected on the <i>MV Example</i> are considered as invasive marine organisms in Australia or New Zealand, and all species detected are cosmopolitan with established ranges including Australia and New Zealand. However, biofouling in some locations exceeded the “short-stay” threshold in the NZ CRMS and thus required remedial cleaning. Table 3 provides a summary of areas on the hull of the <i>MV Example</i> exceeding the CRMS “short-stay” threshold and requiring remedial cleaning. Figure 2 provides the approximate locations of all areas requiring cleaning.		

³ X, 2018. Inspection of the *MV Example* to determine the level of biofouling and detect Invasive Marine Species. Report for Company X. Report Reference X; Version 1.0.

Table 3. Submerged hull areas of the *MV Example* requiring remedial cleaning following the initial inspection conducted while the vessel was alongside at Singapore Cruise Centre, Singapore on 19 September, 2018.

Hull Location	Description
Waterline	Forward section of waterline (forward of Frame 198), and aft section of waterline (aft of Frame 14) with <5% scattered barnacles and tubeworms in a band extending to approximately 5 m. Secondary biofouling concentrated on weld seams. Transom waterline with <5% scattered barnacles and tubeworms restricted to starboard quarter.
Forward and Aft Draft markers	Biofouling consisting of <5% acorn barnacles and tubeworms associate with forward draft markers extending to depth of 5 m. Draft markers deeper than 5 m with isolated acorn barnacles (<1%).
Overboard pipes	All inspected overboard pipes found to have significant biofouling consisting of 60 – 90% acorn barnacles, tubeworms, bryozoans and hydroids.
Blades of all tunnel thrusters	Thruster propellers with ~50% biofouling consisting of acorn barnacles, tubeworms, hydroids and algae.
Stabilising fin anode	One anode associated with port side stabiliser fin void with ~50% biofouling consisting of tubeworms, hydroids and acorn barnacles.
Rope guards and rope cutter voids (propeller and shaft)	A single juvenile mussel (<i>Mytilus</i> sp.) was detected and removed from a port side rope cutter blade associated with the shaft. Void spaces of shaft and propeller rope guards with 40-60% secondary biofouling consisting primarily of tubeworms with scattered (~5%) acorn barnacles.
Interface surfaces of propellers	Trailing face of propeller boss with 70-90% secondary biofouling composed primarily of tubeworms with scattered acorn barnacles. Rudder facings with ~10% tubeworms with scattered acorn barnacles.
Rudders	Rudder hinges and void spaces with ~60% secondary biofouling consisting of acorn barnacles and tubeworms. Anodes with 20-40% secondary biofouling consisting of acorn barnacles and tubeworms. Non-toxic surfaces associated with top of rudder and rudder stock with 20-60% secondary biofouling. Scattered individual mussels (<i>Mytilus</i> sp.) present around rudder stock of both rudders (~10%).

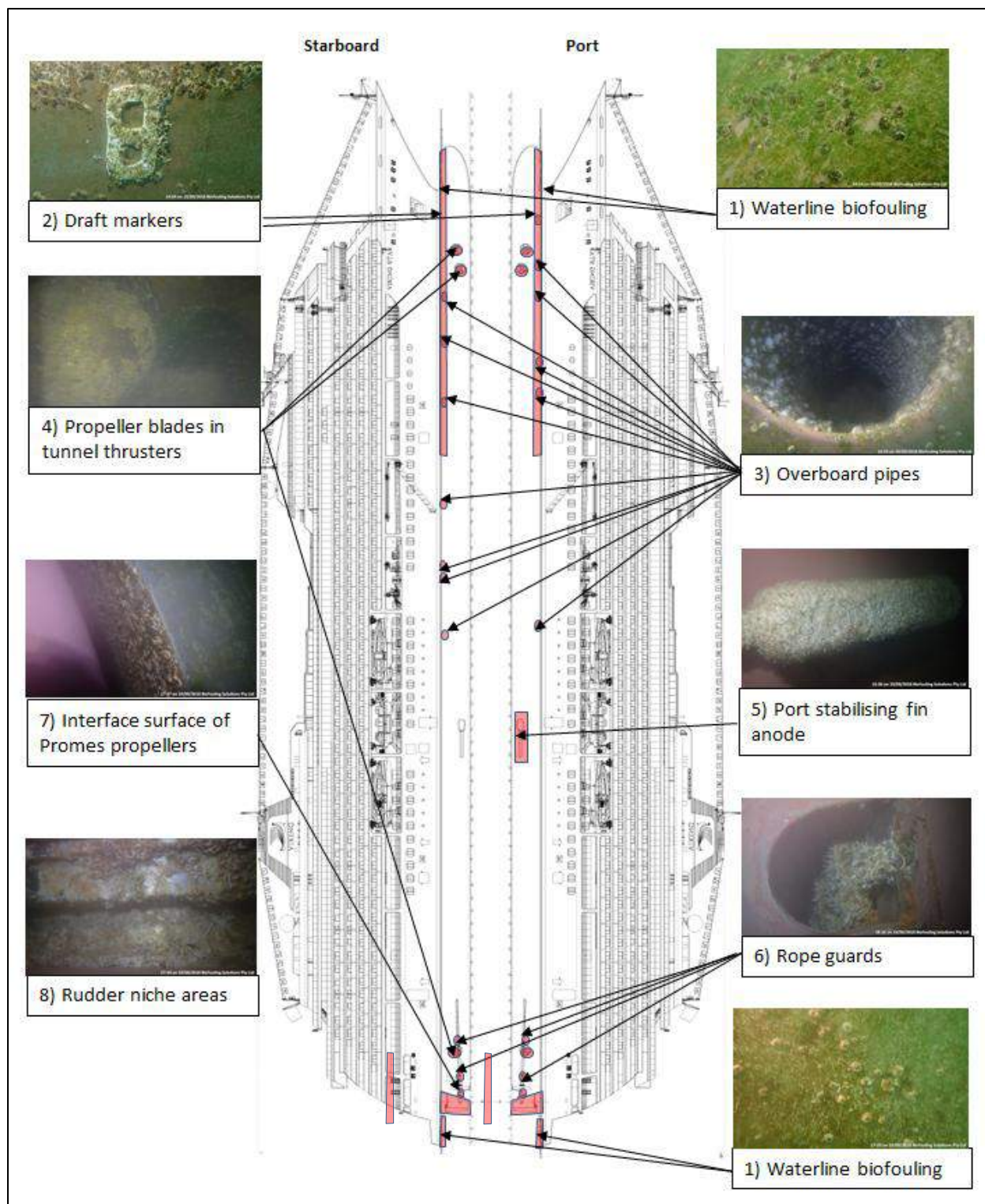


Figure 2 Submerged areas of the *MV Example* inspected while the vessel was alongside at Singapore Cruise Centre, Singapore on 19 September, 2018 that require remedial cleaning, and examples of biofouling detected in excess of CRMS thresholds.

4.2 Cleaning operation 1 of 2: Laem Chabang, Thailand

X oversaw the first of two in-water cleaning operations while the *MV Example* was at anchor in Laem Chabang, Thailand. The goal of this operation was to address the scope of work outlined above, with a key focus on completing brush kart cleaning operations to remove waterline biofouling. A summary of cleaning methodology and outcomes is provided below:

START DATE AND TIME:	13:00 hours, 17 November, 2018	
FINISH DATE AND TIME:	15:30 hours, 17 November, 2018	
DIVE COMPANY:	X	
DIVE VESSEL:	X	
DIVE SUPERVISOR:	X	
DIVERS:	1 diver on SSBA with HP water jet (3000 psi); 1 diver on SCUBA operating BrushKart; 8 divers on SCUBA conducting manual cleaning with hand scrapers.	
DIVING AND INSPECTION EQUIPMENT USED:	☒ Surface supply (SSBA) ☒ SCUBA ☒ HP Water Jet (3000 psi)	☒ Communications and CCTV ☒ Hand-scraper ☒ BrushKart
ALL TARGETED HULL LOCATIONS CLEANED :	☐ Yes	☒ No
TARGETED HULL LOCATIONS CLEANED OR INSPECTED:	Cleaning of rudders and rope guard areas not completed. X inspector oversaw cleaning of all port overboard pipes and verified cleaning was conducted to a high standard. All other areas cleaned by SCUBA divers and not verified.	
RANGE/AVERAGE WATER CLARITY DURING FINAL INSPECTION:	8 m	
TIDAL CURRENTS DURING FINAL INSPECTION:	<1 knots	
INSPECTION RECORDED ON DVD:	☐ Yes	☒ No
COMMENTS: X verified the HP water jet cleaning of 37/39 overboard pipes on the port side, and verified that the brushkart cleaning scope of work was completed save a 10 m section on the port aft section. Despite significant cleaning efforts, available time did not allow for the completion of the entire scope of work and subsequent inspection requirements, and several areas required additional cleaning and inspection to verify that the standards set out in the NZ CRMS were met. Table 4 provides a summary of areas on the hull of the <i>MV Example</i> still exceeding the CRMS “short-stay” threshold and requiring further remedial cleaning at the conclusion of the operation in Laem Chabang, and Figure 3 provides the approximate locations of all areas requiring cleaning.		

Table 4. Submerged hull areas of the *MV Example* requiring remedial cleaning following the initial in-water cleaning operation conducted at Laem Chabang, Thailand on 19 November, 2018.

Hull Location	Description of area to be cleaned and scope of work
Port Waterline	Hand-clean section of waterline biofouling on the port side between Frames 1-10.
Overboard pipes	Inspect and verify standard of cleaning on starboard side overboard pipes.
Tunnel thrusters	Inspect and verify standard of cleaning on forward and aft tunnel thruster blades.
Rope guards & rope cutters	Clean void spaces of shaft and propeller rope guards using HP water jet. Inspect and verify standard of cleaning.
Interface surfaces of propellers	Clean trailing face of propeller boss using HP water jet. Inspect and verify standard of cleaning.
Rudders	Clean non-antifouled rudder surfaces. Inspect and verify standard of cleaning.

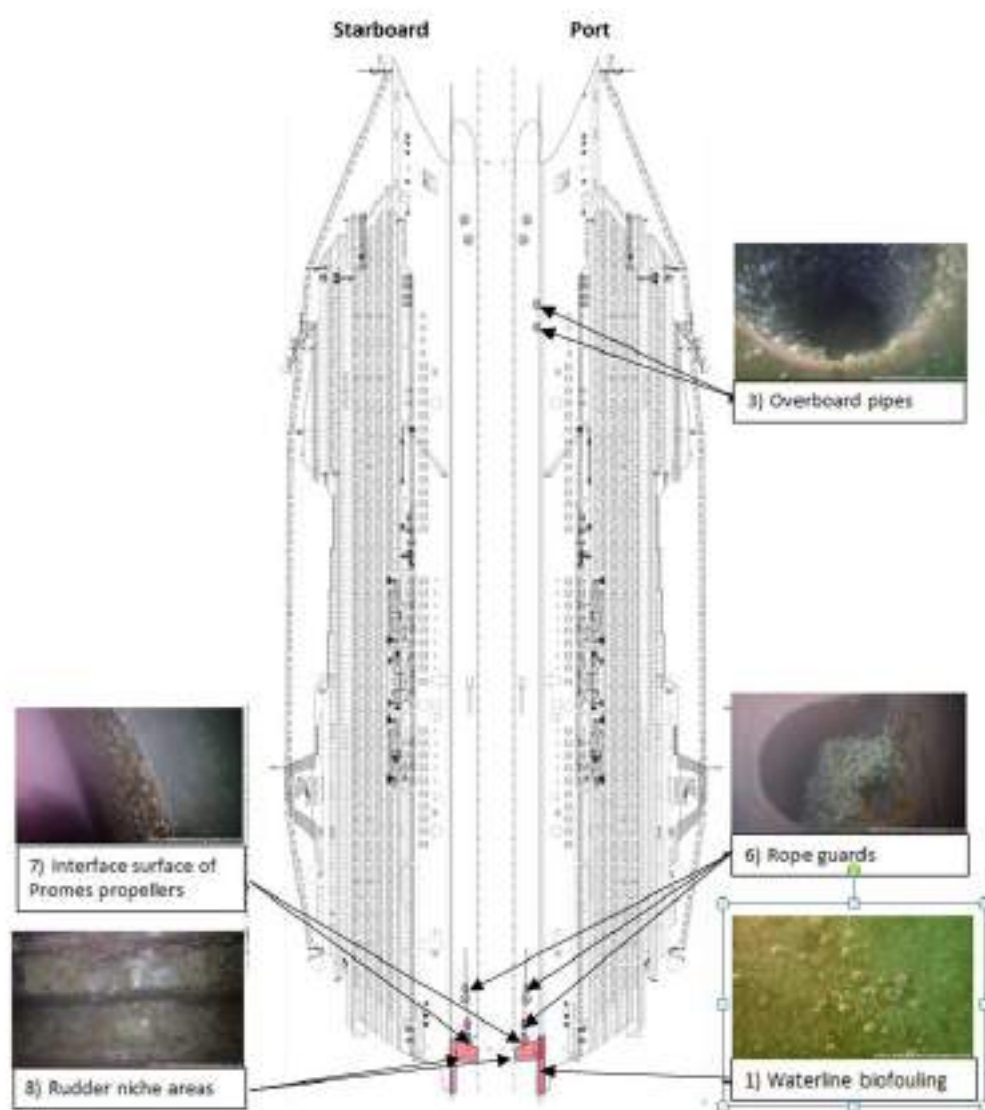


Figure 3 Submerged areas of the *MV Example* requiring remedial cleaning following the initial in-water cleaning operation conducted at Laem Chabang, Thailand on 19 November, 2018.

4.3 Final External Hull Cleaning and Inspection

X oversaw the final in-water cleaning operation and conducted a final biofouling inspection while the *MV Example* was alongside at the Singapore Cruise Centre, Singapore on 19 November, 2018. The final inspection determined that all cleaning was conducted to a high standard and no residual biofouling was detected in excess of the NZ CRMS “short-stay” threshold. A summary of cleaning methodology and inspection outcomes is provided below:

START DATE AND TIME:	11:50 hours, 23 September, 2018	
FINISH DATE AND TIME:	17:00 hours, 23 September, 2018	
DIVE COMPANY:	X	
DIVE VESSEL:	X	
DIVE SUPERVISOR:	X	
DIVER:	Various	
DIVING AND INSPECTION EQUIPMENT USED:	☒ Surface supply (SSBA) ☒ Digital Camera ☒ HP Water Jet (3000 psi)	☒ Communications and CCTV ☒ Hand-scraper
ALL HULL LOCATIONS/NICHE AREAS INSPECTED:	☒ Yes (with previous)	☐ No
RANGE/AVERAGE WATER CLARITY DURING FINAL INSPECTION:	1 m	
TIDAL CURRENTS DURING FINAL INSPECTION:	<1 knots	
INSPECTION RECORDED ON DVD:	☒ Yes	☐ No
COMMENTS: The final initial inspection confirmed that all cleaning operations were completed to a high standard and remaining biofouling in these areas was limited to non-viable calcium carbonate (i.e. acorn barnacle bases and tubeworm scars) with only scattered individual acorn barnacle remaining with densities less than 1% over the general hull and less than 5% in niche areas. Table 5 provides the details of biofouling recorded following remedial cleaning action, indicates compliance with the NZ CRMS and the dates each location was cleaned and inspected. Figure 4 provides a summary of all inspection dive pathways. A full assessment of the cleaning and inspection results is provided in the discussion section below.		

Table 5. The final biofouling inspection results showing the status of biofouling on the *MV Example* following the completion of all in-water cleaning operations. The pooled results of the initial inspection are also included where no remedial cleaning was required. See **Appendix 4** for representative photographs of areas that did not require cleaning, and **Appendix 5** for before and after images showing areas that required remedial cleaning.

Niche Type	No, inspected/ No. present	Date cleaned	Final inspection	Compliant with CRMS	Status after cleaning
Bulbous bow	1/1	17/11/18	23/11/18	Y	Some isolated acorn barnacles (<1%).
Waterline	1/1	17/11/18 23/11/18	23/11/18	Y	Some isolated acorn barnacles (<1%) and diatomaceous slime.
General hull	1/1	N/A	19/09/18 23/11/18	Y	Mostly clean with light biofouling consisting of green algae, diatomaceous slime and scattered acorn barnacles at a density of ~ 1%.
Bilge keels	2/2	N/A	19/09/18	Y	Mostly clean with diatomaceous slime only. Some small areas of damaged paint (<1%).
Draft markers	5/6	17/11/18 23/11/18	23/11/18	Y	All draft markers either hand cleaned or cleaned via brush kart. Clean with no viable biofouling.
Overboard pipes (port)	38/40	17/11/18	17/11/18	Y	Cleaned using HP water jet (3000 psi). Generally clean (<1%) with biofouling limited to barnacle bases.
Overboard pipes (port)	12/30	17/11/18 23/11/18	23/11/18	Y	Cleaned using hand scrapers. Some barnacle bases remaining (~5%) but no viable biofouling.
Sea chest grates	3/6	N/A	19/09/18	Y	Only non-operational sea chests could be inspected. Biofouling restricted to 1-5% acorn barnacles and tubeworms (<1%), generally associated with bolts or areas of damaged paint.
Sea chest boxes	3/6	N/A	19/09/18	Y	Only non-operational sea chests could be inspected and inspection conducted through grates (gratings not removed). Inside of sea chests appeared clean with biofouling mostly restricted to diatomaceous slime. Port aft low sea chest had isolated acorn barnacles inside (<1%).

Niche Type	No, inspected/ No. present	Date cleaned	Final inspection	Compliant with CRMS	Status after cleaning
Sounder equipment	3/3	N/A	19/09/18	Y	Mostly clean with some secondary biofouling mostly associated with bolts and consisting of acorn barnacles (<5%) and tubeworms (<1%).
Tunnel thrusters	3/3	17/11/18	23/11/18	Y	Grating entered by SCUBA divers in Thailand to facilitate cleaning. All propeller surfaces hand scraped with residual biofouling limited to acorn barnacle bases and serpulid tubeworm scars (<5%). Gratings with <1% acorn barnacles and tubeworms mostly associated with damages paint surfaces. Thruster tunnels and anodes were clean with biofouling restricted to diatomaceous slime. Gear housing with scattered acorn barnacles and tubeworms (<5%).
Stabilising fins	2/2	17/11/18	19/09/18 17/11/18	Y	Void space and fin/hinge surfaces mostly clean with biofouling restricted to ~5% acorn barnacles and ~1% tubeworms. Anodes associated with port side stabiliser fin void cleaned using HP water jet.
Rudders	2/2	17/11/18 23/11/18	23/11/18	Y	Plating surfaces of rudders clean with no biofouling. Rudder hinges and void spaces cleaned using HP water jet. Following cleaning, residual biofouling limited to <5% non-viable acorn barnacle bases
Propellers	2/2	N/A	19/09/18	Y	Blades mostly clean with no biofouling. Scattered dead acorn barnacles associated with base of blades (<5%).
Interface surfaces of 'Promes' propellers	2/2	23/11/18	23/11/18	Y	Cleaned using HP water jet. Following cleaning, residual biofouling limited to <5% non-viable serpulid tubeworm bases
Propeller shaft and brackets	2/2	23/11/18	23/11/18	Y	Shaft and bracket surfaces clean with no biofouling.

Niche Type	No, inspected/ No. present	Date cleaned	Final inspection	Compliant with CRMS	Status after cleaning
Rope guards and rope cutter voids (propeller and shaft)	10/10	23/11/18	23/11/18	Y	Cleaned using HP water jet. Following cleaning, residual biofouling limited to <5% biofouling in inaccessible areas consisting of serpulid tubeworms and acorn barnacles.
Dry-Dock Support Strips (DDSS)	various	19/09/18	19/09/18	Y	DDSS mostly clean with only diatomaceous slime and occasional examples of either tubeworms or acorn barnacles (<1%).
Ballast plugs	various	19/09/18	19/09/18	Y	Ballast plugs mostly clean with occasional examples of either tubeworms or acorn barnacles (<1%).

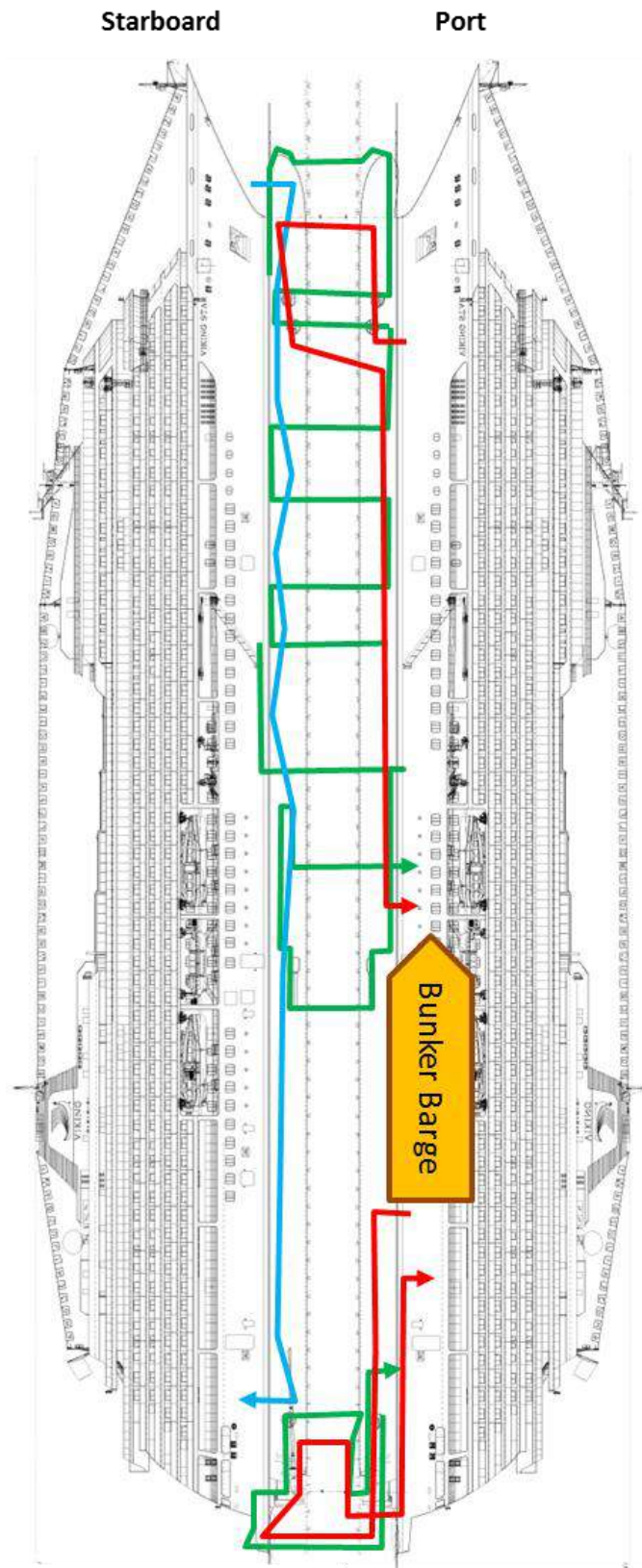


Figure 4 Summary of inspection dive paths undertaken in the initial inspection on 19 September, 2018 (—); the cleaning operation conducted on 17 November, 2018 (—); and the final IMS inspection conducted on 23 November, 2018 (—). The position of the bunkering barge during both diving operations in Singapore is also shown.

4.4 *MV Example* Topside Immersible Equipment Inspection

Company X has indicated that various items of topside equipment are stored on board the decks of the *MV Example*. While most of this equipment rarely contacts the sea, the tenders stored upon the main deck (port and starboard) have been used previously, and will be used again to transport passengers between the vessel and shore during tourist operations in Queensland in 2019 (i.e. prior to visiting New Zealand). To assess the level of biofouling entrained upon immersible topside equipment, X conducted an inspection of all topside equipment accessible on the main deck while the *MV Example* was alongside at the Singapore Cruise Centre, Singapore on 23 November, 2018. A summary of inspection outcomes is provided below:

INSPECTION STARTED:	16:00 hours	19 November, 2018
INSPECTION COMPLETED:	16:30 hours	19 November, 2018
ALL REQUESTED EQUIPMENT INSPECTED:	☒ Yes ☐ No	See Table 6 for details.
EQUIPMENT INSPECTED:	Life rafts, Life boats, Work boats, Life rings, Tenders	
COMMENTS: Company X requested that X inspect all immersible equipment that might be used prior to entry into New Zealand water to assess compliance with the NZ CRMS biofouling thresholds. The inspection found all equipment in very good order with no biofouling detected. Table 6 provides a summary of the inspection results.		

Table 6. Biofouling inspection results for topside equipment aboard the *MV Example*. See **Appendix 6** for representative photographs.

Niche Type	No, inspected / No. present	Biofouling Level (C/P/S/T) ⁴	IMS detected (Y/N)	Comments	Cleaned (Y/N)
Life rafts	various	C	N	Clean and in as-new condition. No entrained biofouling.	N
Life Boats	4/4	C	N	Clean and in as-new condition. No entrained biofouling.	N
Tender / Life Boats	2/2	C	N	Clean. No entrained biofouling.	N
Work Boats	2/2	C	N	Clean. No entrained biofouling.	N
Life rings	various	C	N	Clean and in as-new condition. No entrained biofouling.	N

⁴ Level of Biofouling: C = Clear, P = Primary, S = Secondary, T = Tertiary. For further details see Appendix 3.

4.5 Limitations and Declarations

The following limitations of the inspection are acknowledged. Quality assurance declarations are provided in the Appendices as appropriate.

Limitation or Declaration	Acknowledged
The inspection was unable to detect microscopic or juvenile life stages of IMS of concern or general biofouling.	☒ Yes ☐ No ☐ N/A
It is possible that IMS of concern or biofouling species could have settled on the vessel during or immediately after the inspection or in-water clean.	☒ Yes ☐ No ☐ N/A
If excessive biofouling was present, the inspection may not have been able to detect certain IMS, particularly if they were in very low abundance.	☐ Yes ☐ No ☒ N/A
The inspection only focussed on target IMS of concern and biofouling thresholds on the submerged external hull. Hence, biofouling or IMS of concern in ballast tanks was not included in this Scope of Works. It is therefore up to the vessel to independently comply with relevant ballast water management requirements.	☒ Yes ☐ No ☐ N/A
The client requested that suspected IMS detected during the inspection be confirmed by taxonomic experts or genetic confirmation.	☐ Yes ☐ No ☒ N/A
This report considers information received from third parties including the client. X proceed on the basis that the information supplied is accurate and complete, and has not sought to independently verify such information. X assumes no responsibility for any reporting conclusions or subsequent actions influenced by the provision of inaccurate or incomplete information to X.	☒ Yes ☐ No ☐ N/A
The X Diving Quality Assurance Declaration has been signed by the commercial diving company responsible for conducting all the in-water inspection and/or cleaning operations (see Appendix 7).	☒ Yes ☐ No ☐ N/A

5. DISCUSSION

An initial in-water biofouling inspection of the *MV Example* conducted while alongside at Singapore Cruise Centre, Singapore on 19 September, 2018, determined that the hull was in good condition with biofouling on the general hull typically consisting of diatomaceous slime and a waterline band of green algae. Secondary biofouling on the hull plating was largely restricted to the forward sections of the waterline (forward of Frame 198), the aft sections of waterline (aft of Frame 14) and the starboard quarter of the transom. This secondary biofouling was associated with a band of green algae extending to approximately 2.5 m depth and consisted of scattered barnacles (*Amphibalanus reticulatus* <5%) and tubeworms (*Hydroides elegans* <1%), with biofouling concentrated on weld seams and the waterline. Some scattered secondary biofouling consisting of individual acorn barnacles and tubeworms (<1%) was detected over other areas of the hull plating. A thorough inspection of niche areas determined that biofouling in niche areas was typically restricted to secondary biofouling composed of acorn barnacles (c.f. *A. reticulatus*), tubeworms (*H. elegans*), hydroids and bryozoans. In most cases biofouling in niche areas represented less than 5% coverage. However, biofouling associated with overboard pipes, draft markers (forward and aft), tunnel thruster propeller blades, anodes associated with the port stabiliser fin, rope guard voids and rudder stock areas exceeded this level. Localised secondary biofouling in some restricted niche areas represented up to 90% coverage. Tertiary biofouling composed of scattered individual mussels (*Mytilus galloprovincialis*) was restricted to the rudder stock areas and a single individual associated with a port rope cutter blade. The X Inspector achieved a high level of confidence that IMS of concern would have been detected if present, and that biofouling in excess of the CRMS thresholds were accurately described by the inspection.

No macroscopic IMS of concern listed in regulatory requirements pertaining to Australian state legislation were detected in association with the *MV Example*. Thus, at the conclusion of the initial inspection, the vessel was determined to be compliant with the regulatory obligations of the Northern Territory, Queensland, New South Wales, Victoria and Tasmania. However, it should be recognised that microscopic life-history stages may have been present and not detected, and that individuals of IMS of concern may settle on the vessel's submerged hull subsequent to the initial inspection.

Based on the biofouling observed in the initial inspection, X designed a scope of work to remove all viable biofouling in excess of the NZ CRMS short-stay threshold. This threshold is an appropriate target as NZ MPI have indicated that if a Craft Risk Management Plan (CRMP) is prepared and approved for a vessel, the Ministry for Primary Industries (MPI) are willing to accept a "short-stay threshold" as a reasonable compliance goal for cruise operators. This scope of work was completed through two dive operations overseen by a X inspector with the first occurring in Laem Chabang, Thailand on 17 November, 2018, and the second occurring while alongside at the Singapore Cruise Centre on 23 November, 2018. Divers used a

range of techniques to remove biofouling including a diver operated brush kart targeting waterline biofouling, high pressure water jets operated at 3000 psi to remove biofouling from complex or curved niche areas such as rope guard voids and overboard pipes, and hand scrapers to remove biofouling from accessible niche areas. Tunnel thrusters were accessed via SCUBA divers through the fixed grating and all surfaces were hand cleaned to remove biofouling.

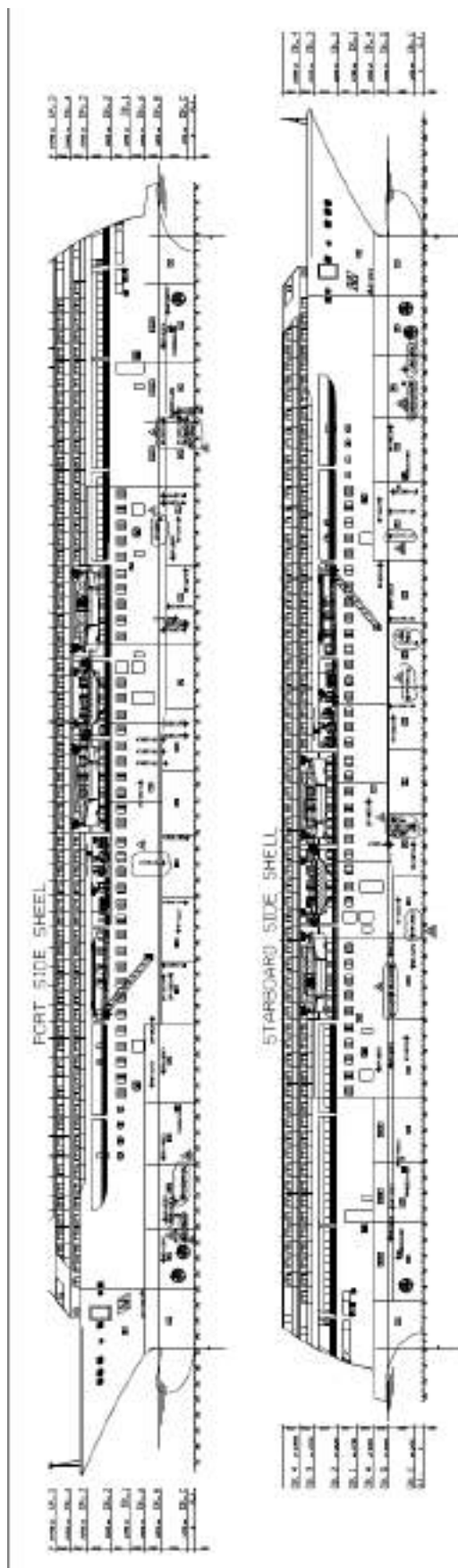
The final dive operation in Singapore on 23 November, 2018 included a full biofouling inspection to ensure that all cleaning was conducted to a high standard, and that no biofouling in excess of the NZ CRMS short-stay threshold persisted on the submerged surfaces and immersible equipment of the *MV Example*. The X Inspector also confirmed that no macroscopic IMS of concern listed in Australian state regulation had recruited to the hull since the initial inspection was conducted on 23 September, 2018. As such, at the conclusion of the final IMS inspection the *MV Example* was found to be in compliance with all regulatory obligations associated with biofouling during the documented 2018/2019 cruise season.

X acknowledges that microscopic life-history stages may have been present in association with the *MV Example* and not detected, and that individuals of IMS of concern or other biofouling species may settle on the vessel's submerged hull subsequent to the inspection. To minimise the chance of infection with IMS of concern, the *MV Example* should limit exposure to IMS propagules by minimizing time spent in coastal waters prior to entering Australian waters. Considering that the *MV Example* is not scheduled to enter New Zealand waters until 21 January, 2019 (~58 days from the final inspection) there is a considerable potential for additional biofouling to accumulate on the submerged surfaces prior to entry. Due to regulations on in-water cleaning in Australia, no further opportunities to remove biofouling prior to entry to New Zealand are available and X considers that the measures described here represent best practice in reducing biosecurity risks to New Zealand associated with biofouling on the *MV Example* to as low as reasonably practicable (ALARP).

6. REFERENCES

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Appendix 1 General Arrangement



Appendix 2 Antifouling Certificate



CERTIFIED COPY

Certificate no: **VEN 1890876**
Page 1 of 2

International Anti-Fouling System Certificate

(This certificate shall be supplemented by a Record of Anti-Fouling Systems)
Issued under the International Convention on the Control of Harmful Anti-Fouling Systems on Ships
under the authority of the Government of the Kingdom of Norway by Lloyd's Register EMEA.

When a Certificate has been previously issued this Certificate replaces the certificate dated **Not Applicable**

	Particulars of ship
Name of ship	
Port of registry	
Distinctive numbers or letters	
Gross tonnage	
IMO number	

An anti-fouling system controlled under Annex 1 has not been applied during or after construction of this ship. ☒

An anti-fouling system controlled under Annex 1 has been applied on this ship previously, but has been removed by _____ or _____. ☐

An anti-fouling system controlled under Annex 1 has been applied on this ship previously, but has been covered with a sealer coat applied by _____. ☐

An anti-fouling system controlled under Annex 1 was applied on this ship prior to _____ but will be removed or covered with a sealer coat prior to _____. ☐

This is to certify that

- the ship has been surveyed in accordance with regulation 1 of Annex 4 to the Convention; and
- the survey shows that the anti-fouling system on the ship complies with the applicable requirements of Annex 1 to the Convention

Issued at **Venice** on **07 June 2018**
Date of completion of the survey on which this certificate is issued **27 September 2017**



A. Solano
Surveyor to Lloyd's Register EMEA
a member of the Lloyd's Register group

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Form 1425 (2/01/05)

CERTIFIED COPY

Certificate no:
Page 2 of 2

VEN 1690876

Record of Anti-Fouling Systems

This record shall be permanently attached to the International Anti-Fouling System Certificate

	Particulars of ship
	Name of ship
	Distinctive numbers or letters
	IMO number
Type(s) of anti-fouling system(s) used	Details of anti-fouling system applied Biocidal
Date(s) of application of anti-fouling system(s)	25 September 2017
Name(s) of company(ies) and facility(ies)/location(s) where applied	Fincantieri S.p.A. - Ancona Shipyard
Name(s) of anti-fouling system manufacturer(s)	Jotun Paints (Europe) Ltd., Stather Road, Flakborough, Scunthorpe, UK,
Name(s) and colour(s) of anti-fouling system(s)	SeaQuantum Ultra 5 - Dark Red
Active ingredient(s) and their Chemical Abstract Service Registry Number(s) (CAS Number(s))	CD - Cuprous Oxide - 1317-39-1 CP - Cuprous Pyrrhione - 14815-37-8
Type(s) of sealer coat, if applicable	Not Applicable
Name(s) and colour(s) of sealer coat applied, if applicable	Not Applicable
Date of application of sealer coat	Not Applicable

This is to certify that this record is correct in all respects.

Issued at: **Venice**

On: **07 June 2018**

Lloyd's Register EMEA
A member of the Lloyd's Register Group



A. Solano
Venice Office
A. Solano

Surveyor to Lloyd's Register EMEA

a member of the Lloyd's Register group.

Form 1425 (2015-08)

Appendix 3 Levels of Biofouling

The Commonwealth of Australia has devised a simplistic, but pragmatic method of assessing the likelihood of vessels or infrastructure being infested by IMS of concern. It relates likelihood of an IMS being present to the level of complexity in the biofouling assemblage. Levels of biofouling tend to increase the longer a vessel or structure remains submerged in seawater. Three levels of biofouling are identified on the basis of the organisms present. The likelihood of IMS of concern being present amongst biofouling increases markedly as the assemblage develops from primary to secondary and finally to tertiary levels.

The colonisation and accumulation of biofouling on surfaces submerged in seawater depend on a series of complex processes. Primary biofouling of submerged surfaces begins with biochemical and bacterial conditioning the moment a vessel or structure is immersed in seawater. This is followed by colonisation by diatoms, protozoans and multi-cellular organisms. This conditioning and colonisation of surfaces by microscopic organisms provides an ideal substrate for settlement and growth of more visible organisms such as fine filamentous algae, (some of which are resistant to the toxic biocides contained in anti-fouling coatings). In turn their establishment provides a suitable, (but not necessarily critical) substrate for settlement of the secondary biofouling organisms which are most commonly encountered during inspections. Secondary biofouling is more likely to develop into tertiary biofouling in niche areas of vessels or structures that are protected from strong hydrodynamic forces, when vessels remain stationary for long periods of time, or as the in-service period for the vessel increases.

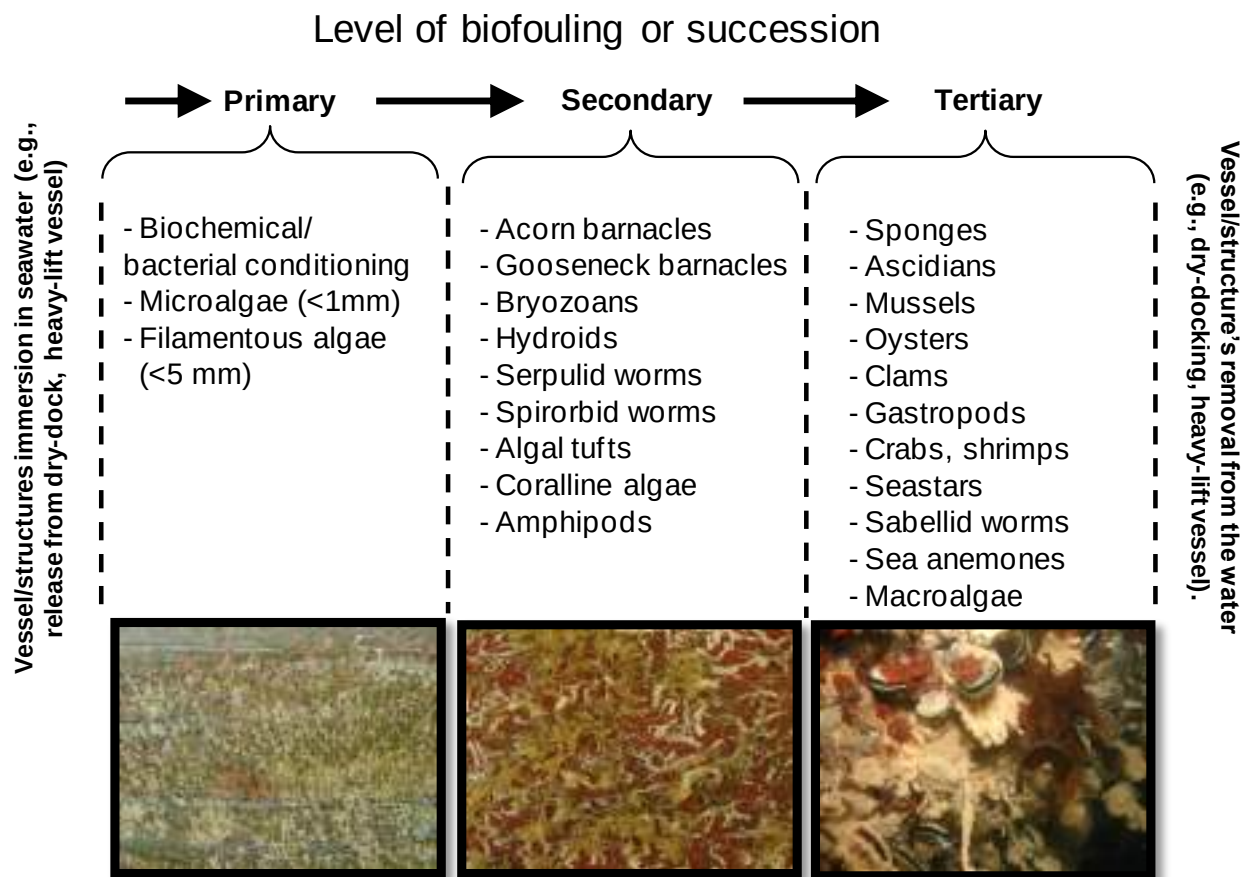


Figure 3. The simplistic, but pragmatic approach to assessing IMS likelihood using three levels of biofouling.

Appendix 4 External Hull Inspection Photos

This appendix shows representative images of hull and niche areas verified as compliant with the New Zealand CRMS “Short-stay” threshold during the initial IMS inspection conducted on 19 September, 2018. While these areas were reinspected during the final IMS inspection and confirmed to fall within the threshold, no cleaning was conducted. Further photos can be made available on DVD upon request.



Plate 1 Clean hull below 5m. Forward of Frame 198, port.



Plate 2 Clean hull below 5m. Forward of Frame 198, starboard.



Plate 3 Clean hull below 5m. Forward of Frame 198, starboard.



Plate 4 Band of green algae without secondary biofouling. Starboard waterline aft of Frame 198.



Plate 5 Port side forward draft markers deeper than 5m with no secondary biofouling.



Plate 6 Starboard side forward draft markers deeper than 5m with no secondary biofouling.



Plate 7 Depth sounder with <5% secondary biofouling restricted to bolts. Forward amidships



Plate 8 Speed log with <5% secondary biofouling. Forward amidships.



Plate 9 Flat bottom shell plating. Forward amidships.



Plate 10 Flat bottom shell plating. Amidships



Plate 11 Dry Dock Support Strip, amidships.



Plate 12 Dry Dock Support Strip, amidships.



Plate 13 Dry Dock Support Strip, amidships.



Plate 14 Amidships waterline algae free of secondary biofouling. Port.



Plate 15 Amidships waterline algae free of secondary biofouling. Starboard.

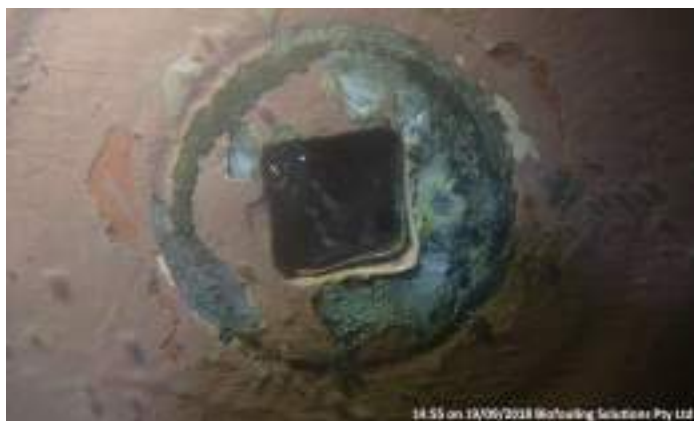


Plate 16 Ballast plug, flat bottom amidships.



Plate 17 Ballast plug, flat bottom amidships.



Plate 18 Ballast plug, flat bottom amidships.



Plate 19 Starboard side bilge keel, leading edge.



Plate 20 Starboard side bilge keel, upper surface.



Plate 21 Starboard side bilge keel, lower surface.



Plate 22 Port side bilge keel, leading edge.



Plate 23 port side bilge keel, upper surface.



Plate 24 Bow thruster #1 grating. Starboard side.



Plate 25 Bow thruster #1 grating. Starboard side.



Plate 26 Bow thruster #1 grating. Port side showing secondary biofouling <5%.



Plate 27 Bow thruster #1. Gear housing viewed through grating.



Plate 28 Bow thruster #1. Mounting support viewed through grating.



Plate 29 Bow thruster #1. Anode and tunnel wall viewed through grating.



Plate 30 Bow thruster #2 grating. Port side.



Plate 31 Bow thruster #2 grating. Port side.



Plate 32 Bow thruster #2 grating. Starboard side.



Plate 33 Bow thruster #2. Anode and prop viewed through grating.



Plate 34 Bow thruster #2. Gear housing viewed through grating.



Plate 35 Bow thruster #2. Mounting support viewed through grating.



Plate 36 Bow thruster #2. Anode and tunnel wall viewed through grating.



Plate 37 Port side stabilising fin. Inspected in retracted position.



Plate 38 Port side stabilising fin. Inspected in retracted position.



Plate 39 Port side stabilising fin. Inspected in retracted position.



Plate 40 Port side stabilising fin. Inspected in retracted position.



Plate 41 Clean anode on port side stabilising fin. Inspected in retracted position.



Plate 42 Starboard side stabilising fin. Inspected in retracted position.



Plate 43 Starboard side stabilising fin. Inspected in retracted position.



Plate 44 Clean anode on Starboard side stabilising fin. Inspected in retracted position.



Plate 45 Starboard side stabilising fin. Inspected in retracted position.



Plate 46 Forward high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard.



Plate 47 Forward high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard.



Plate 48 Forward high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard.



Plate 49 Forward high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard.



Plate 50 Amidships high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard.



Plate 51 Amidships high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard



Plate 52 Aft low sea chest grating. Port.



Plate 53 Aft low sea chest grating. Port.

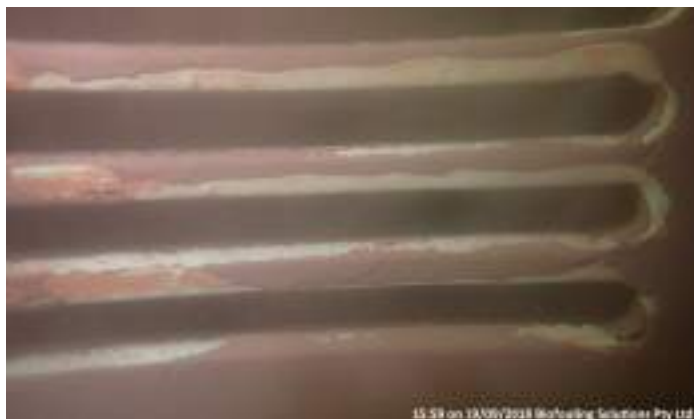


Plate 54 Aft low sea chest grating. Port.



Plate 55 Aft low sea chest grating. Port.



Plate 56 Aft low sea chest grating. Port.



Plate 57 Aft low sea chest grating with <1% secondary biofouling. Port.

Appendix 5 Final External Hull Inspection Photos showing biofouling before and after cleaning

Photos in this appendix are representative of the locations and equipment inspected. Further photos can be made available on DVD upon request.



Plate 1 PRIOR TO CLEANING: Band of acorn barnacles >1%. Forward of Frame 198, port, waterline to 5m.



Plate 2 PRIOR TO CLEANING: Acorn barnacles >1% associated with weld seam. Forward of Frame 198, port, waterline to 5m.



Plate 3 AFTER CLEANING: Forward port waterline after cleaning. Forward of Frame 198.



Plate 4 AFTER CLEANING: Forward port waterline weld-seam after cleaning. Forward of Frame 198.



Plate 5 AFTER CLEANING: Residual acorn barnacles on forward port waterline weld-seam with coverage less than 1%. Forward of Frame 198.

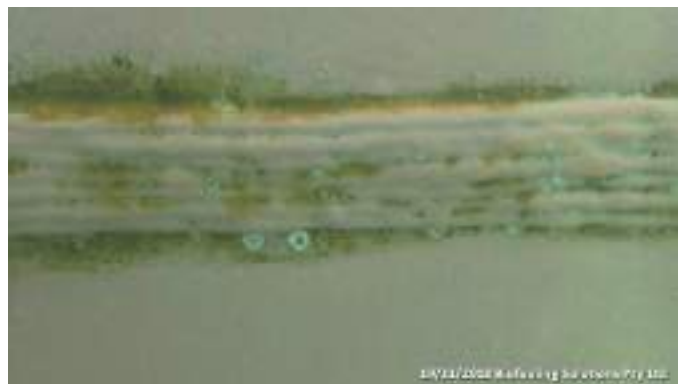


Plate 6 AFTER CLEANING: Residual acorn barnacles on forward port waterline weld-seam with coverage less than 1%. Forward of Frame 198.



Plate 7 PRIOR TO CLEANING Acorn barnacles and tubeworms >1% associated with weld seam. Forward of Frame 198, starboard, waterline.



Plate 8 PRIOR TO CLEANING: Band of acorn barnacles >1%. Forward of Frame 198, starboard, waterline.



Plate 9 AFTER CLEANING: Forward starboard waterline after cleaning with residual algae. Forward of Frame 198.

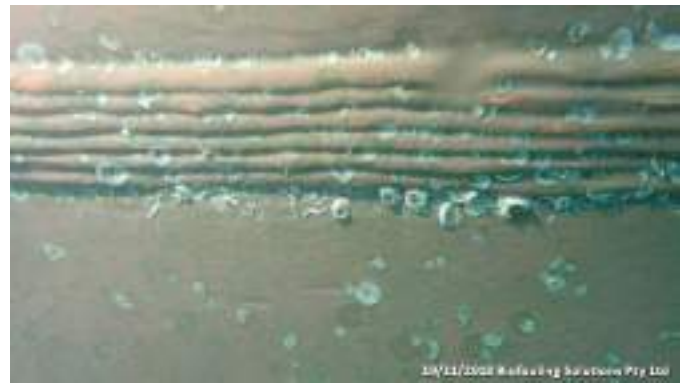


Plate 10 AFTER CLEANING: Residual acorn barnacle bases on forward starboard waterline weld-seam with coverage of intact barnacles less than 1%. Forward of Frame 198.



Plate 11 AFTER CLEANING: Forward starboard waterline weld-seam after cleaning. Forward of Frame 198.



Plate 12 AFTER CLEANING: Residual acorn barnacles on forward starboard waterline weld-seam with coverage less than 1%. Forward of Frame 198.



Plate 13 PRIOR TO CLEANING Port aft shell plating with secondary biofouling >5% associated with weld seam (aft of Frame 14).



Plate 14 PRIOR TO CLEANING Port aft shell plating with secondary biofouling >1% associated with weld seam (aft of Frame 14).



Plate 15 AFTER CLEANING: Aft port waterline after cleaning with residual algae. Aft of Frame 14.



Plate 16 AFTER CLEANING: Aft port waterline after cleaning showing weld seam. Aft of Frame 14.



Plate 17 AFTER CLEANING: Aft port waterline after cleaning showing residual barnacles associated with weld seam (<1%). Aft of Frame 14.



Plate 18 AFTER CLEANING: Aft port waterline after cleaning showing residual barnacles associated with weld seam (<1%). Aft of Frame 14.



Plate 19 PRIOR TO CLEANING Starboard aft shell plating with secondary biofouling >1% associated with weld seam (aft of Frame 14).



Plate 20 PRIOR TO CLEANING Starboard aft shell plating with secondary biofouling >1% associated with weld seam (aft of Frame 14).



Plate 21 AFTER CLEANING: Aft starboard waterline after cleaning. Aft of Frame 14.



Plate 22 AFTER CLEANING: Aft starboard waterline after cleaning showing residual barnacles (<1%). Aft of Frame 14.



Plate 23 AFTER CLEANING: Aft starboard waterline after cleaning showing weld seam. Aft of Frame 14.



Plate 24 AFTER CLEANING: Aft starboard waterline after cleaning showing residual non-viable barnacle bases associated with weld seam (<1%). Aft of Frame 14.



Plate 25 PRIOR TO CLEANING: Port side forward draft markers showing secondary biofouling > 5%.



Plate 26 PRIOR TO CLEANING: Starboard side forward draft markers showing secondary biofouling > 5%.



Plate 27 AFTER CLEANING: Port side forward draft.



Plate 28 AFTER CLEANING: Port side forward draft markers.



Plate 29 AFTER CLEANING: Starboard side forward draft markers.



Plate 30 AFTER CLEANING: Starboard side forward draft markers.



Plate 31 PRIOR TO CLEANING: Bow thruster #1. Prop viewed through grating. Divers observed ~50% Secondary biofouling on prop and boss.



Plate 32 AFTER CLEANING: Bow thruster #1. Prop viewed through grating after cleaning.

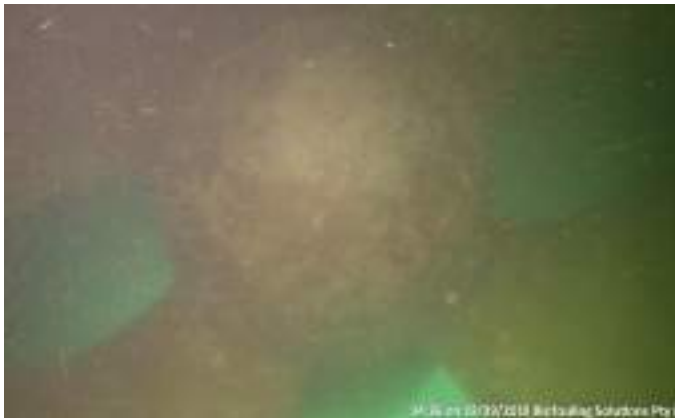


Plate 33 PRIOR TO CLEANING: Bow thruster #2. Propeller viewed through grating. Divers observed ~50% Secondary biofouling on prop and boss.



Plate 34 AFTER CLEANING: Bow thruster #2. Prop viewed through grating after cleaning.



Plate 35 AFTER CLEANING: Aft tunnel thruster boss after cleaning.



Plate 36 AFTER CLEANING: Aft tunnel thruster propeller blade after cleaning.



Plate 37 PRIOR TO CLEANING: Example of biofouling >5% on overboard pipes.



Plate 38 PRIOR TO CLEANING: Example of biofouling >5% on overboard pipes.



Plate 39 AFTER CLEANING: Example of overboard pipes after cleaning. Biofouling limited to non-viable acorn barnacles bases.



Plate 40 AFTER CLEANING: Example of overboard pipes after cleaning. Biofouling limited to non-viable acorn barnacles bases.



Plate 41 AFTER CLEANING: Example of overboard pipes after cleaning. Biofouling limited to non-viable acorn barnacles bases.



Plate 42 AFTER CLEANING: Example of overboard pipes after cleaning. Biofouling limited to non-viable acorn barnacles bases.



Plate 43 PRIOR TO CLEANING: Example of biofouling >5% within rope-guard void spaces



Plate 44 PRIOR TO CLEANING: Example of biofouling >5% within rope-guard void spaces



Plate 45 AFTER CLEANING: Example of rope guard void after cleaning.



Plate 46 AFTER CLEANING: Example of rope guard void after cleaning.



Plate 47 AFTER CLEANING: Example of rope guard void after cleaning.



Plate 48 AFTER CLEANING: Example of rope guard void after cleaning.



Plate 49 PRIOR TO CLEANING: Example of biofouling >5% on rudder anodes.



Plate 50 PRIOR TO CLEANING: Example of biofouling >5% on rudder anodes.



Plate 51 AFTER CLEANING: Example of rudder anode after cleaning.



Plate 52 AFTER CLEANING: Example of rudder anode after cleaning.



Plate 53 AFTER CLEANING: Example of rudder anode after cleaning.



Plate 54 AFTER CLEANING: Example of rudder anode after cleaning.



Plate 55 PRIOR TO CLEANING: Top of port rudder stock with >5% secondary biofouling.



Plate 56 PRIOR TO CLEANING: example of >5% secondary biofouling and scattered tertiary biofouling (mussels) associated with rudders.



Plate 57 AFTER CLEANING: Example of non-antifouled rudder surfaces after cleaning.



Plate 58 AFTER CLEANING: Example of non-antifouled rudder surfaces after cleaning.



Plate 59 AFTER CLEANING: Example of non-antifouled rudder surfaces after cleaning.



Plate 60 AFTER CLEANING: Example of non-antifouled rudder surfaces after cleaning.



Plate 61 PRIOR TO CLEANING: Port Promes propeller showing interface surface with >5% secondary biofouling.



Plate 62 PRIOR TO CLEANING: Example of >5% secondary biofouling associated with Promes propeller.



Plate 63 AFTER CLEANING: Example of rudder leading face of surface after cleaning.



Plate 64 AFTER CLEANING: Example of propeller boss surface after cleaning.

Appendix 6 Top ide equipment of the *MV Example*

Photos in this appendix are representative of the locations and equipment inspected. Further photos can be made available on DVD upon request.



Plate 1 Port side life rafts



Plate 2 Port side work boat



Plate 3 Port side work boat

Plate 4 Port side tender / lifeboat #1



Plate 5 Port side tender / lifeboat #1



Plate 6 Port side tender / lifeboat #1



Plate 7 Port side tender / lifeboat #3



Plate 8 Port side tender / lifeboat #3



Plate 9 Port side tender / lifeboat #3



Plate 10 Port side tender / lifeboat #5



Plate 11 Port side tender / lifeboat #5



Plate 12 Port side tender / lifeboat #5



Plate 13 Starboard side life rafts



Plate 14 Starboard side work boat



Plate 15 Starboard side tender / lifeboat #2

Plate 16 Starboard side tender / lifeboat #2



Plate 17 Starboard side tender / lifeboat #2



Plate 18 Starboard side tender / lifeboat #4



Plate 19 Starboard side tender / lifeboat #4



Plate 20 Starboard side tender / lifeboat #4



Plate 21 Starboard side tender / lifeboat #6



Plate 22 Starboard side tender / lifeboat #6



Plate 23 Starboard side tender / lifeboat #6



Plate 24 Starboard side tender / lifeboat #6

Appendix 7 X Diver Quality Assurance Declaration

Form No.

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

Vessel/infrastructure's name: _____ BFS Job #: _____
Location: SINGAPORE CRUISE CENTRE Starting Date: 19-08-17

The Dive Supervisor declares that they understand and agree to the following separation of roles:

1. _____ supplies an Invasive Marine Species (IMS) inspector to:
 - A. Assess compliance of the vessel/infrastructure with the appropriate IMS management requirements. In order to assess compliance they may need to do the following:
 - B. Provide guidance on the standards and requirements of the inspection and/or the cleaning required for the vessel/infrastructure. This may involve requests, directions or instructions being provided to Divers via the Dive Supervisor.
 - C. Request the collection of biological samples and/or photographs from specified areas.
2. The Dive Supervisor is responsible for all aspects of diving safety, the coordination of divers and has the final decision on all aspects of the dive team's professional activities. The Dive Supervisor must:
 - A. Conduct appropriate risk assessments to ensure the safety of divers and diving operations.
 - B. Ensure procedures, equipment and dive team members are safe and appropriate for all aspects of the inspection and/or cleaning operations.
 - C. Be present at all times while Divers are undertaking the inspection or cleaning, and conduct communication with Divers underwater. In addition,
 - D. If the Dive Supervisor objects to the IMS Inspector's requests, directions or instructions they must promptly discuss all relevant issues with the IMS Inspector to safeguard the Divers.

The Dive Supervisor declares that they understand the IMS inspection and/or clean must achieve the standards stipulated by the following regulations and/or biofouling management requirements:

- ☐ Chevron Australia Pty Ltd's Barrow Island – Marine Vessels Procedure
☐ Woodside Energy Ltd's Invasive Marine Species Management Plan
☐ Western Australian Department of Fisheries *Fish Resources Management Act 1994*
☒ Other: NZ CRMS

Initials: Dive Supervisor: AS Contractor: PM

Form No.

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

Treatment standard required:

- ☒ Level of Biofouling (i.e. Remove Primary, Secondary and/or Tertiary Levels of Biofouling)
☐ Remove Suspected Invasive Marine Species of Concern

Diving activity required:

- ☒ Inspection
☐ In-water clean

The under-signed declares that they and their divers took all reasonable steps to conduct the inspection and were willing and able to:

- ☒ Follow directions given by the _____ inspector
☒ Achieve the standards requested by the _____ inspector
☐ Conduct in-water cleaning treatments to ensure the complete removal of macroscopic secondary or tertiary biofouling and/or suspected Invasive Marine Species of Concern from the following locations:

N/A

The under-signed declares that diver's did not leave any areas where biofouling exceeded the aforementioned standard except for:

N/A

Initials: Dive Supervisor:

AD

Contractor:

PNZ

Form No.

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

These could not be cleaned because:

N/A

Additional Comments:

INSPECTION TO NZ CRMS EXAMINING LEVEL OF
FOLLING.

Initials: Dive Supervisor:

NA

Contractor:

PNZ

Form No

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

The under-signed also declares that their divers conducted the final closed circuit television survey in such a manner that the _____ IMS inspector was able to accurately verify that the cleaning had achieved the aforementioned standard.

Commercial Diving Company: _____

Name: _____ Official capacity: Dive Supv.

Signed: [Signature] Date: 19/09/18

Contractor: _____

Name: _____ Official capacity: _____

Signed: _____ Date: _____

Name: _____ Official capacity: _____

Signed: _____ Date: 19-09-18

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

Vessel/infrastructure's name: _____ BFS Job #: _____

Location: LAEM CHABANG, THAILAND Starting Date: 17/11/18

The Dive Supervisor declares that they understand and agree to the following separation of roles:

1. _____ supplies an Invasive Marine Species (IMS) inspector to:
 - A. Assess compliance of the vessel/infrastructure with the appropriate IMS management requirements. In order to assess compliance they may need to do the following:
 - B. Provide guidance on the standards and requirements of the inspection and/or the cleaning required for the vessel/infrastructure. This may involve requests, directions or instructions being provided to Divers via the Dive Supervisor.
 - C. Request the collection of biological samples and/or photographs from specified areas.
2. The Dive Supervisor is responsible for all aspects of diving safety, the coordination of divers and has the final decision on all aspects of the dive team's professional activities. The Dive Supervisor must:
 - A. Conduct appropriate risk assessments to ensure the safety of divers and diving operations.
 - B. Ensure procedures, equipment and dive team members are safe and appropriate for all aspects of the inspection and/or cleaning operations.
 - C. Be present at all times while Divers are undertaking the inspection or cleaning, and conduct communication with Divers underwater. In addition,
 - D. If the Dive Supervisor objects to the IMS Inspector's requests, directions or instructions they must promptly discuss all relevant issues with the IMS Inspector to safeguard the Divers.

The Dive Supervisor declares that they understand the IMS inspection and/or clean must achieve the standards stipulated by the following regulations and/or biofouling management requirements:

- ☐ Chevron Australia Pty Ltd's Barrow Island – Marine Vessels Procedure
- ☐ Woodside Energy Ltd's Invasive Marine Species Management Plan
- ☐ Western Australian Department of Fisheries Fish Resources Management Act 1994
- ☒ Other: NZ CRMS

Initials: Dive Supervisor: KD Contractor: _____

27/11

Form No.

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

Treatment standard required:

- ☒ Level of Biofouling (i.e. Remove Primary, Secondary and/or Tertiary Levels of Biofouling)
☐ Remove Suspected Invasive Marine Species of Concern

Diving activity required:

- ☐ Inspection
☒ In-water clean

The under-signed declares that they and their divers took all reasonable steps to conduct the inspection and were willing and able to:

- ☒ Follow directions given by the _____ inspector
☒ Achieve the standards requested by the _____ inspector
☒ Conduct in-water cleaning treatments to ensure the complete removal of macroscopic secondary or tertiary biofouling and/or suspected Invasive Marine Species of Concern from the following locations:

REMOVE BIOFOULING AS PER SCOPE OF WORK

- BRUSHKART ON BOW / STEERN
- OVERBOARD PIPES
- RUDDER / PROPS

The under-signed declares that diver's did not leave any areas where biofouling exceeded the aforementioned standard except for:

- ROPE GUARDS
- PROPELLOR HUBS
- RUDDER PORTLET

Initials: Dive Supervisor: _____ Contractor: _____



Page 2 of 4

Form No.

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

These could not be cleaned because:

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Additional Comments:

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Initials: Dive Supervisor: _____ Contractor: _____

Form No.

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

The under-signed also declares that their divers conducted the final closed circuit television survey in such a manner that the _____ IMS inspector was able to accurately verify that the cleaning had achieved the aforementioned standard.

Commercial Diving Company: _____

Name: _____ Official capacity: SENIOR SUPERINTENDENT

Signed: [Signature] Date: NOV 17, 2018

Contractor: _____

Name: _____ Official capacity: _____

Signed: _____ Date: _____

Name: _____ Official capacity: _____

Signed: _____ Date: 17-11-18

Form No.

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

Vessel/infrastructure's name: _____

Location: SINGAPORE CRUISE CENTRE, Starting Date: 23-11-18
SINGAPORE

The Dive Supervisor declares that they understand and agree to the following separation of roles:

1. _____ supplies an Invasive Marine Species (IMS) inspector to:
 - A. Assess compliance of the vessel/infrastructure with the appropriate IMS management requirements. In order to assess compliance they may need to do the following:
 - B. Provide guidance on the standards and requirements of the inspection and/or the cleaning required for the vessel/infrastructure. This may involve requests, directions or instructions being provided to Divers via the Dive Supervisor.
 - C. Request the collection of biological samples and/or photographs from specified areas.
2. The Dive Supervisor is responsible for all aspects of diving safety, the coordination of divers and has the final decision on all aspects of the dive team's professional activities. The Dive Supervisor must:
 - A. Conduct appropriate risk assessments to ensure the safety of divers and diving operations.
 - B. Ensure procedures, equipment and dive team members are safe and appropriate for all aspects of the inspection and/or cleaning operations.
 - C. Be present at all times while Divers are undertaking the inspection or cleaning, and conduct communication with Divers underwater. In addition,
 - D. If the Dive Supervisor objects to the IMS Inspector's requests, directions or instructions they must promptly discuss all relevant issues with the IMS Inspector to safeguard the Divers.

The Dive Supervisor declares that they understand the IMS inspection and/or clean must achieve the standards stipulated by the following regulations and/or biofouling management requirements:

- ☐ Chevron Australia Pty Ltd's Barrow Island – Marine Vessels Procedure
☐ Woodside Energy Ltd's Invasive Marine Species Management Plan
☐ Western Australian Department of Fisheries *Fish Resources Management Act 1994*
☒ Other: NZ CRMS

Initials: Dive Supervisor: AS Contractor: PNB



Form No.

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

Treatment standard required:

- ☒ Level of Biofouling (i.e. Remove Primary, Secondary and/or Tertiary Levels of Biofouling)
- ☒ Remove Suspected Invasive Marine Species of Concern

Diving activity required:

- ☒ Inspection
- ☒ In-water clean

The under-signed declares that they and their divers took all reasonable steps to conduct the inspection and were willing and able to:

- ☒ Follow directions given by the _____ inspector
- ☒ Achieve the standards requested by the _____ inspector
- ☒ Conduct in-water cleaning treatments to ensure the complete removal of macroscopic secondary or tertiary biofouling and/or suspected Invasive Marine Species of Concern from the following locations:

CLEANING: RUDDERS, RIFEGUARDS, PT STERN WATERLINE

VERIFICATION/INSPECTION: ALL AREAS

The under-signed declares that diver's did not leave any areas where biofouling exceeded the aforementioned standard except for:

ALL AREAS CLEANED/INSPECTED

Initials: Dive Supervisor: TA Contractor: _____

PNB

Form No.

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

These could not be cleaned because:

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Additional Comments:

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Initials: Dive Supervisor: _____ Contractor: _____

Handwritten initials PM

Form No.

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

The under-signed also declares that their divers conducted the final closed circuit television survey in such a manner that the _____ IMS inspector was able to accurately verify that the cleaning had achieved the aforementioned standard.

Commercial Diving Company: SSG

Name: T.A. Jones Official capacity: DIVE SUPERVISOR

Signed: [Signature] Date: 23/11/18

Contractor: _____

Name: _____ Official capacity: _____

Signed: _____ Date: _____

Name: _____ Official capacity: _____

Signed: _____ Date: 23-11-18